Week 7 Week 7 Add one challenge — paper completion route (no device) · paper route answers

Teacher only. Keep this sheet away from pupil copies.

GRID 1, Hazard. when green flag clicked / go to x: 0 y: -100 / forever / then indented inside forever: glide 2 secs to x: 0 y: 100 / glide 2 secs to x: 0 y: -100. Two spare rows stay empty. The two glides must both be inside forever. If they are written below forever instead of inside it, ask what forever is holding.

GRID 2, Player contact rule. when green flag clicked / forever / indented: if <touching [Hazard]?> then / indented again: go to x: -180 y: -120. It must be headed Player. If a pupil heads it Hazard, that is the classroom bug, not a slip: go to Grid 7 and teach it.

GRID 3, Player reset. when green flag clicked / go to x: -180 y: -120.

GRID 4, the four arrow scripts. when [right arrow] key pressed then change x by 10. when [left arrow] key pressed then change x by -10. when [up arrow] key pressed then change y by 10. when [down arrow] key pressed then change y by -10. The minus sign on left and down is the thing to check.

GRID 5, wall rule. when green flag clicked / forever / if <touching [Maze]?> then / go to x: -180 y: -120. It is built the same shape as Grid 2. Say so out loud: the same shape, a different sensing block, a separate stack.

GRID 6, win response. when green flag clicked / say [ ] / go to x: -180 y: -120 / wait until <touching [Finish]?> / say [Made it!]. The empty say must be first. Ask what it is for. Answer: it clears the old bubble when the game restarts.

GRID 7, the bug card. The fault: the contact rule is built on Hazard and it checks touching [Player]?. On contact it moves Hazard to (-180, -120) and Player stays where it is. The repair grid, headed Player, must read: when green flag clicked / forever / if <touching [Hazard]?> then / go to x: -180 y: -120. Then one line through the Hazard card, because that script is removed. The printed fault card itself is never rewritten.

TRACE 1 answers. Row 1: nothing has moved yet, blank is correct. Row 2: time 0, Hazard (0, -100). Row 3: time 2, (0, 100). Row 4: time 4, (0, -100). Row 5: time 6, (0, 100). Row 6: time 8, (0, -100).

TRACE 2 answers, the clock. Step 0: 0.0 s, (0, -100), yes. Step 1: 0.5 s, (0, -50), yes. Step 2: 1.0 s, (0, 0), no. Step 3: 1.5 s, (0, 50), no. Step 4: 2.0 s, (0, 100), no. Step 5: 2.5 s, (0, 50), no. Step 6: 3.0 s, (0, 0), no. Step 7: 3.5 s, (0, -50), yes. Step 8: 4.0 s, (0, -100), yes, and this is the same as step 0, so the cycle repeats.

TRACE 3 answers, the arrows. Row 1: nothing yet, blank is correct. Row 2: (-180, -120). Row 3: (-180, -110). Row 4: (-180, -100). Row 5: (-170, -100). Row 6: (-160, -100). Row 7: (-160, -110). No wall is touched at any row.

SHARED SETUP a. Two clock steps is one second. The Hazard reaches (0, 0) and Player is on (0, 0). Same crossing, so touching. Player returns to (-180, -120). This is the pack's own published expected result.

SHARED SETUP b. The Hazard reaches (0, 0). Player remains on (30, 0). The gap is 3 small squares, and the rule says two crossings apart is not touching. Nothing happens. This matters: the feature does not stop every route.

SHARED SETUP c. After 4 steps the Hazard reaches (0, 100). After 4 more it returns to (0, -100). Then it repeats. Ask which block makes it repeat. Answer: forever.

SHARED SETUP d. The empty say clears the message. Both sprites go back to their positions. Player on the wall square (-60, 0) is touching the wall, so it returns to (-180, -120). Player on Finish (180, 120) gives one new Made it! The earlier rules still work.

BEFORE AND AFTER answers. Row 1, Week 6: Player stays on (0, 0), nothing happens. Row 1, Week 7: at step 2 the Hazard is on (0, 0), they are touching, Player returns to (-180, -120). Row 2, Week 6: Player stays on (30, 0). Row 2, Week 7: Player stays on (30, 0), the Hazard passes 3 squares away. Row 2 giving the same answer twice is correct and is worth pointing out. Row 3: accept any wording that names a new player action, such as I must watch the Hazard, I must wait, I must go round the side, I must time my crossing. Do not accept it looks different.

COMPARE TWO SETTINGS answers. glide 2 secs to each end: 10 small squares in one second, 1 second in the low band on a visit, 3 seconds clear between visits. glide 4 secs to each end: 5 small squares in one second, 2 seconds in the low band on a visit, 6 seconds clear between visits. The teaching point in the pupil's own words: the slow Hazard sits on the crossing for twice as long, and my crossing takes the same time either way, so slower is not automatically fairer. Both answers are acceptable choices. The unacceptable answer is a choice with no test result written beside it.

TRACE 4 answers, the bug. Row 2: after 2 steps the Hazard is on (0, 0) and Player is on (0, 0), touching. Row 3: the go to is on Hazard, so Hazard goes to (-180, -120) and Player stays on (0, 0). Row 4: on the next clock step the glide carries on, so the Hazard reappears on its own line at (0, 50). Player was never returned. The feature has no effect on the player. That is the whole lesson of the bug.

MAZE answers. Safe start (-180, -120). Finish (180, 120). The Hazard line is x 0 from y -100 to y 100. The low band is x 0 from y -100 to y -50, five small squares tall. A working route is (-180, -120) up to (-180, 120), right to (30, 120), down to (30, -120), right to (180, -120), up to (180, 120). That is 108 presses, so about 22 clock steps at the paper play rule. The route does cross the Hazard's line, at (0, 120) on the top run, but that crossing is 2 small squares above the top end of the journey, so it is never touching. On the way down at x 30 the closest approach is 3 small squares, also not touching. A pupil who runs this route and never gets caught has not made a mistake. They have found a usable route, which is what the design sheet asks for.

UNSEEN SETUP U1 answer. Player on (0, 110). At step 4 the Hazard reaches (0, 100). One crossing apart, so touching. Player returns to (-180, -120). Nothing happens at steps 1, 2 or 3.

UNSEEN SETUP U2 answer. Player on (30, -110). No contact at any of the 8 steps. Player stays on (30, -110). The closest the Hazard gets is 3 small squares, at step 0 and again at step 8.

UNSEEN SETUP U3 answer. Player on (0, -110). Touching straight away, before the clock moves at all, because the Hazard begins on (0, -100), one crossing away. Player returns to (-180, -120). This one catches a pupil who has learned that contact happens after two steps rather than learning the rule.

WHAT CORRECT LOOKS LIKE WHEN A PUPIL POINTS. On Grid 2, correct is pointing at the Player heading, then at touching [Hazard]?, then at go to. On the maze, correct is putting a finger on (0, -100), then tracing up the line to (0, 100), then back down. For the before and after table, correct is pointing at (0, 0) for the Week 6 answer and then at (-180, -120) for the Week 7 answer. Pointing at the right two places in the right order is a full answer. Write down what the pupil pointed at.

WHAT CORRECT LOOKS LIKE WHEN A PUPIL DRAWS. A straight line at x 0 between the two marked ends, with an arrow at each end, is a correct journey. An arrow curving back on itself is also correct if the pupil says it repeats. A drawn Hazard of any shape is correct provided the pupil says how wide it is and it leaves a route to Finish. For contact, a drawn arrow from the meeting point back to the safe start is a correct answer to what happens.

WHAT CORRECT LOOKS LIKE WHEN A PUPIL SAYS IT. Scribe the exact words. For outcome 5 the words that count are the ones naming a new player action. Watch the Hazard, wait for it to go up, go round the side, time it, all count. It looks different does not count, and the follow-up question is: what must the player do now that they did not do before?

IF THE PUPIL IS NOT FINISHED. Record what is done, date it, and agree extra time. Do not backfill missing rows from this answer list.

What this week evidences

AQA 71638 Programming with Scratch (unit 6), Level One. Outcome 5: added a feature that changes gameplay. The pack's teacher guide states the evidence: "Observe the pupil creating the feature and explaining what the player must now do. Record a before-and-after test. Decoration alone does not meet the outcome." Evidence for the outcome is a summary sheet completed by the teacher from individual observation. The centre's AQA summary sheet remains the official record and is completed by staff. The statements below are a local scaffold only. The route the pupil took is recorded in the teacher guide, never on the AQA summary sheet.

Known limit of the paper route

Five things are genuinely not evidenced on paper, and staff should not claim them. FIRST, no Scratch sprite is created. Pupil check 1 says "I created a new sprite and named it myself". On paper the pupil designs, draws, names and places a Hazard, and the teacher can observe every part of that. What is not observed is the pupil using the Scratch sprite pane. If a centre reads outcome 5 as requiring the sprite to be made inside Scratch, the paper route does not supply that one act, and the pupil should repeat only that step when a device is next available. SECOND, and this is the real difference between the two routes, no program is ever executed by a computer. On paper the pupil is the interpreter. The teacher observes the pupil applying its own rule correctly and consistently, including moving the Player counter back to (-180, -120) on contact. What is not observed is a built program producing that consequence by itself. This is exactly why the unseen setup above is not optional: without it, a pupil who copies the printed expected results has demonstrated nothing, and the observation would not hold. With it, the observation does hold, because the pupil has to apply the rule to a setup they have not seen an answer for. THIRD, no file exists. The Exit step on the screen route is save, reload, run one check. That has no paper equivalent and must not be ticked. The paper equivalent is dated pages filed and initialled, and it evidences filing, not saving. FOURTH, continuous motion is not seen.

The clock strip shows the same start, the same turning point and the same return in half-second steps, and running it twice round shows the journey repeats. The pupil has not watched a glide. This does not affect outcome 5, which asks what the player must now do, but nobody should claim the pupil watched the Hazard move. FIFTH, the paper play rule, five small squares of Player to one clock step of Hazard, is a fair-play convention chosen so both move ten small squares a second. In Scratch a held arrow key repeats faster than that. The paper game is therefore harder than the screen game at the same settings, and the paper timings are not a measurement of Scratch. One thing does hold exactly, and is worth stating: the paper touching rule and the screen touching rule agree at every grid crossing. Same crossing or a neighbouring crossing, corner to corner included, is touching. Two crossings apart is not. This was checked against the model's own arithmetic at every position on the lattice, so paper and screen give the same answer to every setup in these tables.